

Sai Prabhu Lagisetty

lagisettsaiprabhu@gmail.com | +1-913-238-4680 | [LinkedIn](#) | [GitHub](#) | Work Authorization:F1-OPT

Professional Summary

Backend Software Engineer with 2+ years of experience delivering customer-obsessed solutions in Java, Python, and Flask. Proven track record of reducing system latency by 40%, automating 60% of manual workflows, and leading cross-functional teams of 15+ engineers. Specialized in building scalable, fault-tolerant applications.

Education

Master of Science in Computer Science | University of Central Missouri, Kansas | GPA: 3.6 /4.0

August 2023 – May 2025

Relevant Coursework: Advanced Algorithms, Machine Learning, Cloud Computing, Distributed Systems, Advanced Database Systems, Compiler Design and Construction, Stat Found Data Science & AI ,Big Data Analytics ,Artificial Intelligence.

Technical Skills & Tools

- **Languages:** Java, Python, JavaScript, C#, SQL
- **Frameworks:** Spring Boot, ASP.NET, Flask, .NET
- **Cloud & DevOps:** AWS (EC2, S3, IAM), CI/CD, OAuth 2.0
- **Databases:** MySQL, Oracle, MongoDB
- **Tools:** Git, Elasticsearch, Postman, MuleSoft
- **Machine Learning:** Scikit-learn, Pandas, NLP
- **Practices:** Agile, Performance Tuning, Security Implementation

Professional Experience

Software Engineer Volunteer, America Scores Bay Area

June 2025-Present | Remote, USA

Tech: Python, pytest, Flask, GitHub, API Testing, MuleSoft, Workbench, Postman

- **Owned end-to-end API reliability framework** serving 10,000+ youth program participants, reducing system downtime by 85% and improving user experience satisfaction scores from 3.2/5 to 4.7/5 through proactive monitoring and automated failover mechanisms
- **Accelerated release velocity by 78%** - transformed release cycles from 14 days to 3 days by architecting comprehensive pytest-based automated testing suite, achieving 99% deployment confidence and eliminating 95% of post-release hotfixes
- **Drove 30% program expansion** by developing ML-powered resource allocation engine using Python and scikit-learn, reducing manual scheduling overhead by 25% while enabling data-driven decisions that expanded program reach to 3,000+ additional participants
- **Established 99.9% uptime standard** across distributed microservices architecture by implementing AI-powered workflow automation and real-time performance monitoring, reducing incident response time from 45 minutes to 8 minutes

Software Engineer-Program Analyst, Cognizant Technology Services

October 2021 – July 2023 | Hyderabad, India

Tech: Java, Spring Boot, MySQL, OAuth 2.0, AWS, Dotnet, PLSQL, OAuth

- **Architected and delivered enterprise-scale backend applications** using Java, Spring Boot, and microservices architecture, serving 10,000+ concurrent users with 99.5% uptime and achieving 40% performance improvement through systematic code optimization and database tuning
- **Redesigned critical ticketing system infrastructure** implementing RESTful APIs and event-driven microservices architecture, resulting in 30% faster resolution times and improved SLA adherence across 500+ daily support tickets through intelligent routing algorithms and load balancing
- **Led technical transformation of 15-engineer team** - designed comprehensive system design curriculum that accelerated team productivity by 40% and reduced new hire onboarding from 4 weeks to 2 weeks, resulting in \$50K+ cost savings annually
- **Delivered 25% ticket backlog reduction by building** Remedy Acknowledgment microservice using C# and MySQL, maintaining 99.5% SLA compliance while processing 2,500+ weekly tickets through automated workflow optimization
- **Streamlined hiring efficiency by 20%** - implemented REST API-driven candidate tracking system managing \$2M+ annual hiring budget, reducing administrative overhead by 15% and improving time-to-hire from 45 days to 36 days

Software Developer intern, Cognizant Technology Services

July 2021 – October 2021 | Hyderabad, India

Tech: SQL, Oracle, Python

- **Optimized production systems handling 50K+ daily transactions** - enhanced complex SQL queries achieving 50% execution time improvement through advanced indexing and query restructuring, processing \$5M+ in daily transaction volume
- **Built student information system serving 5K+ users** - improved data retrieval speed by 35% using relational database optimization and caching strategies, reducing average page load time from 2.3 seconds to 1.5 seconds

Projects

E-Book Management System with ML-Powered Sentiment Analysis

Tech: Python, Flask, MySQL, Scikit-learn, Pandas, Selenium, HTML/CSS/JS

- **Architected full-stack e-book management platform** serving 500+ concurrent users with web scraping (Selenium) that processes 1,000+ book entries daily, maintaining 99% uptime and driving 40% increase in content-discovery click-through rates through intelligent caching and database optimization
- **Developed real-time ML-powered sentiment classification engine** using Scikit-learn and NLP models that analyzed 50,000+ user reviews with 87% accuracy, cutting infrastructure costs by 60% while powering personalized recommendation algorithms that boosted user engagement by 35%
- **Enhanced end-to-end system performance** through advanced database indexing and query restructuring, decreasing page load times from 3.2 seconds to 0.8 seconds and API response times by 70%, while handling 25% increase in traffic with zero downtime

Cross-Platform GitHub Repository Issue Analytics & Export Pipeline

Tech: Python, GitHub API, Data Processing, Excel/CSV Export, Pandas, Multi-Platform Deployment

- **Developed cross-platform GitHub issue extraction system** that runs seamlessly on Windows, macOS, and Linux environments, processing 5,000+ issues per repository through robust GitHub REST API integration with built-in rate limiting and error handling, minimizing project management overhead by 70% across diverse development teams
- **Engineered intelligent data pipeline with incremental append functionality** that updates existing Excel files without data loss, handling 50+ repositories simultaneously while maintaining 99.5% data accuracy and preserving historical issue tracking across multiple sprint cycles and project phases
- **Established due diligence framework** that validates data integrity, performs duplicate detection, and ensures compliance with organizational reporting standards, processing repositories with 10,000+ issues while generating audit-ready documentation and maintaining full traceability of data transformations

Certifications

- Microsoft Certified: Azure AI Fundamentals (AI-900)